East West University
Department of Mathematics and Data Science (MDS)
Final Examination (Summer 2026)
Course Title: **Linear Algebra and Complex Variables**
Course Code: **MAT 205** [redacted]
Course Instructor : Md. Ashraful Alam (Alam)

Total Marks: 30

Time: 80 minutes

Answer all the following questions. Figures in the right margin indicate marks.

1. a) How do you define a subspace of a vector space? Examine whether the following subset of $\mathbb{R}^3$ is a subspace of $\mathbb{R}^3$ and justify your answer. [2]
$T = \{(x, y, z) \in \mathbb{R}^3 : x, y, z \in \mathbb{R},\ x^2 + y^2 + z^2 \leq 1\}$.

b) Determine whether the following set of vectors is a basis of $\mathbb{R}^3$ or not. [1]
$\{(1,3-4), (1,4,-3), (2,3,-11)\}$.

c) Diagonalize the following matrix by finding a suitable matrix P: [5]

$$A = \begin{pmatrix} 1 & 0 \\ 6 & -1 \end{pmatrix}.$$

d) Applying Cayley-Hamilton theorem find the inverse of the matrix A, where [2]

$$A = \begin{bmatrix} 2 & 2 & 3 \\ 1 & -1 & 0 \\ -1 & 2 & 1 \end{bmatrix}.$$

2. a) Test whether the transformation defined as follows is linear or not.

$$T: \mathbb{R}^3 \to \mathbb{R}^2 : T(x, y, z) = (z, x + y).$$ [2]

b) Verify whether the function $f(z) = \frac{1}{z}; z \neq 0$ is analytic. [2]

c) For the function $f(z)$, locate and name all the singularities in the finite z-plane, where $f(z) = \frac{z^8+z+2}{(z-1)^3(3z+2)^2}$. [4]

d) Explain the meaning of a non-isolated singularity. Illustrate your answer with suitable examples and distinguish them from isolated singularities. [2

a) State Cauchy's Residue theorem and using this, evaluate $\oint_C \frac{\cos z}{z(z-1)^2} dz$, where C is the circle $|z-1| = \frac{1}{2}$. Finally, verify the result using Cauchy's Integral formula.

Evaluate $\int_0^{2\pi} \frac{d\theta}{3+2\cos\theta}$, using the method of contour integration.

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